S.no.	Number System	Base/ radix	First digit	Last digit	All digits/characters
1.	Binary	2	0	1	0, 1
2.	Octal	8	0	7	0, 1, 2, 3, 4, 5, 6, 7
3.	Decimal	10	0	9	0, 1, 2, 3, 4, 5, 6, 7, 8, 9
4.	Hexadecimal	16	0	F	0, 1, 2, 3, 4, 5, 6, 7, 8, 9 A, B, C, D, E, F

Relation between decimal, binary, octal & hexadecimal no's :-

Decimal	Binary	Octal	Hexadecimal
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000		8
9	1001		9
10	1010		A
11	1011		B
12	1100		C
13	1101		D
14	1110		E
15	1111		F

1) Binary to Octal Conversion :-

1) Convert $(11110\,1100)_2$ to Octal.

base of binary = 2 $\Big)$ 2^3 { ∵ grouping three digits }
base of Octal = 8 $\Big)$

111	101	100	$\longrightarrow$	$(754)_8$ = Ans
↓	↓	↓		
7	5	4		

2) Octal to Binary Conversion :-

a) Convert $(634)_8$ to binary :-

6	3	4	$\longrightarrow$	$(110\,011\,100)_2$ = Ans
↓	↓	↓		
110	011	100		

b) Convert $(725.63)_8$ to binary.

7	2	5	.	6	3	$\longrightarrow$	$(111\,010\,101.110\,011)_2$ = Ans
↓	↓	↓		↓	↓		
111	010	101		110	011		

Binary to Hexadecimal

base of binary - 2) 2^4 {grouping 4 digits}
base of Hexadec - 16 ↙

1) Convert $(1101\ 1000\ 1001\ 1011)_2$ to Hexadec.

1101 → D
1000 → 8
1001 → 9
1011 → B

$\rightarrow (D89B)_H$

Hexadecimal to Binary

1) Convert $(3FD)_H$ to binary

3 → 0011
F → 1111
D → 1101

$\rightarrow (0011\ 1111\ 1101)_2$ =Ans

2) Convert $(5A9.B4)_H$ to Binary

5 → 0101
A → 1010
9 → 1001
.
B → 1011
4 → 0100

$\rightarrow (0101\ 1010\ 1001 . 1011\ 0100)_2$ =Ans

Octal to Hexadecimal Conversion

→ First Convert Octal to its binary equivalent
→ Then, Convert binary no. to its Hexadecimal equivalent.

1) Convert $(615)_8$ to hexadecimal.

⇒ Octal to Binary

6 → 110, 1 → 001, 5 → 101 ⟶ $(110\ 001\ 101)_2$

⇒ Binary to Hexadecimal.

Binary ⟶ $(1\ 1\ 0\ 0\ 0\ 1\ 1\ 0\ 1)_2$ { start pairing with LSB }

(2nd) (1st)

0001 → 1, 1000 → 8, 1101 → D ⟶ $(18D)_H$.

= Ans

Hexadecimal to Octal Conversion

⇒ Convert Hexadec. to its Binary equivalent.

⇒ Convert binary no. to its Octal equivalent.

1) Convert $(25B)_H$ to its octal equivalent.

⇒ Hexadec. to binary →

2	5	B
↓	↓	↓
0010	0101	1011

Binary → $(0010\ 0101\ 1011)_2$

⇒ Binary to Octal →

Binary →

0 0 1	0 0 1	0 1 1	0 1 1
4th	3rd	2nd	1st
↓	↓	↓	↓
1	1	3	3

Octal no - $(1133)_8$ = Ans

Converting any radix to decimal

In general, no's can be represented as under -

$$N = A_{n-1} r^{n-1} + A_{n-2} r^{n-2} + ___ + A_1 r^1 + A_0 r^0 + A_{-1} r^{-1} + A_{-2} r^{-2} + ___ C_{-m} r^{-m}$$

N = No. in decimal
A = digit
r = radix / base
n = no. of digits in the integer portion of no.
m = no. of digits in the fractional portion of no.

Q1 Convert $(1101.1)_2$ to its decimal equivalent :-

$$N = 1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0 + 1\times2^{-1}$$
$$= 8 + 4 + 0 + 1 + 0.5$$
$$= (13.5)_{10} = \underline{\text{Ans}}$$

Que Convert $(475.25)_8$ to its decimal equivalent.

$$N = 4\times8^2 + 7\times8^1 + 5\times8^0 + 2\times8^{-1} + 5\times8^{-2}$$
$$= 256 + 56 + 5 + 0.25 + 0.078125 = (317.32813)_{10} = \text{Ans}$$

Que Convert $(9B2.1A)_H$ to its decimal equivalent.

$$N = 9\times16^2 + B(11)\times16^1 + 2\times16^0 + 1\times16^{-1} + A(10)\times16^{-2}$$
$$= 2304 + 176 + 2 + 0.0625 + 0.039$$
$$= (2482.1)_{10} = \text{Ans}$$

Que Convert $(3102.12)_4$ to its decimal equivalent.

$$N = 3\times4^3 + 1\times4^2 + 0\times4^1 + 2\times4^0 + 1\times4^{-1} + 2\times4^{-2}$$
$$= 192 + 16 + 0 + 2 + 0.25 + 0.125$$
$$= (210.375)_{10} = \text{Ans}$$

Conversion of decimal no's to any radix no.

Que. Convert dec. no. 24.6 to binary no.

⇒ Integer part = 24 fractional part = 0.6

	Q	R	
2	24		LSB.
2	12	0	
2	6	0	
2	3	0	
2	1	1	
	0	1	(MSB)

Q = Quotient R = Remainder.

Binary equivalent of integer part = $(11000)_2$

Fraction		radix		result	recorded carries	
0.6	×	2	=	1.2	1	(MSB)
0.2	×	2	=	0.4	0	
0.4	×	2	=	0.8	0	
0.8	×	2	=	1.6	1	
0.6	×	2	=	1.2.	1	LSB.

$\therefore (24.6)_{10} \rightarrow (11000.10011)_2$ = Ans

Que. Convert decimal no. (35.45) to octal no.

	Q	R.	
8	35		LSB
8	4	3	↑
	0	4	MSB

$(43)_8$.

0.45×8	$= 3.6$	3	(MSB)
0.6×8	$= 4.8$	4	↓
0.8×8	$= 6.4$	6	
0.4×8	$= 3.2$	3	
0.2×8	$= 1.6$	1	(LSB).

$\therefore (35.45)_{10} \longrightarrow (43.34631)_8$ = Ans

Que Convert dec. no. (22.64) to hexadecimal no.

16	22		LSB
16	1	6	↑
	0	1	(MSB)

0.64×16	$= 10.24$	$=$	10 (A)
0.24×16	$= 3.84$	$=$	3
0.84×16	$= 13.44$	$=$	13 (D)
0.44×16	$= 7.04$	$=$	7.

↓

$\therefore (22.64)_{10} \longrightarrow (16.A3D7)_{16}$ = Ans.